

Shape-Changing Water Droplets on Conductive Patterns Printed on Paper

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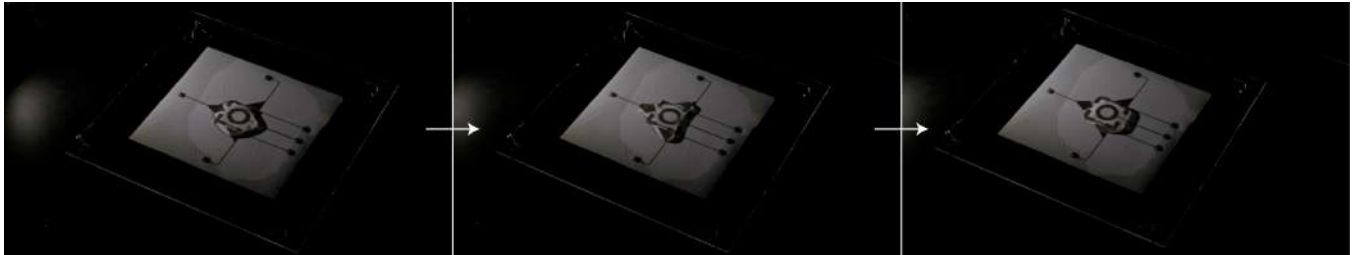


Figure 1: The behavior of a dynamically shape-changing water droplet on a single sheet with printed conductive patterns: (Left) circular, (Middle) triangular, and (Right) square.

Abstract

Droplets offer rich potential as a medium for physical interaction. While droplet transportation has been extensively explored using techniques such as electrowetting, changing the shape of droplets remains technically challenging with conventional methods. If shape-changing droplets become possible, they could open up novel expressive possibilities. In this demonstration, we present a novel approach for enabling dynamic shape-changing of water droplets on conductive patterns printed on paper. Unlike conventional electrowetting-based devices, this work does not require a high-density electrode array and allows easy customization of electrode shapes. It enables shape deformation by controlling the wettability of droplets. By switching the activated electrodes, multiple patterns of shape-changing droplets can be dynamically realized on a single sheet. In this study, we demonstrate the approach to

shape-changing water droplets and present several design examples.

CCS Concepts

• **Human-centered computing** → Interactive systems and tools; Interaction devices.

Keywords

Programmable Materials, Water Droplet Control, Shape Change, Fabrication Technique, Electrowetting

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1 Introduction

In recent years, the field of Human-Computer Interaction (HCI) has increasingly drawn on materials science to enable dynamic transformations of physical form and behavior [7, 26]. Such control

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over physical properties opens new possibilities for interactions that are more embodied, tangible, and materially expressive.

Among various materials, liquid has recently attracted growing attention as substances whose physical properties can be dynamically altered. It exhibits various characteristics—such as thermal conductivity [8, 9, 16, 20, 31], high density [11, 12, 21, 22, 30], ease of coloring [2, 10, 15, 19, 23], and chemical reactivity [13, 17]—which have inspired diverse interface research and expressions. In particular, droplet manipulation has been widely studied across fields including medicine, biology, chemistry, and materials science. Various digital microfluidic platforms have been developed to automate and enhance the efficiency of small-scale liquid handling [3, 6].

One of the most widely used droplet manipulation techniques is electrowetting [14, 18, 24, 25]. Electrowetting is a phenomenon in which a water droplet placed on a hydrophobic dielectric layer coated over an electrode changes its wettability when a high voltage is applied, allowing the control of its contact angle. Droplet transportation based on this principle has played a pivotal role in enabling high-mobility fluidic interfaces [1, 4, 5, 27–29]. Conventional electrowetting-based devices typically achieve droplet movement by densely arranging arrays of small square electrodes on a flat surface and selectively applying voltage to each electrode. However, while electrowetting excels at transporting droplets, it offers little control over the shape of the droplet itself. If droplet shape transformation were made possible, it could open up novel expressive and interactive possibilities beyond the scope of traditional droplet manipulation techniques.

Motivated by these challenges, we propose a novel approach for enabling dynamic shape-changing of water droplets on conductive patterns printed onto a paper-like sheet. Unlike conventional droplet manipulation techniques such as electrowetting, this work does not require a high-density electrode array and instead supports flexible customization of electrode shapes and layouts. It enables shape-changing droplets by controlling their wettability. Specifically, users can easily print various conductive electrode patterns using a household inkjet printer and commercially available silver nanoparticle ink. When a high voltage is applied to printed electrodes, a droplet is drawn along the shape of that electrode, deforming its contour accordingly. By selectively activating different electrodes, multiple patterns of shape-changing droplets can be dynamically realized on a single sheet. A prior study using inkjet printers and silver nanoparticle ink has addressed dense electrode arrangements by focusing on droplet movement between sparsely arranged electrodes [29], while this study focuses on shape change of water droplets using different electrode geometries. In addition to the droplet movement explored in many existing systems, enabling droplet shape change expands the expressive capacity of droplet-based interfaces, opening new directions for interaction design.

In this study, we demonstrate the approach to shape-changing water droplets and present several design examples.

2 Design and Mechanism of Shape-Changing Water Droplets

2.1 System Design

Fig. 2 shows the system design. From top to bottom, the system consists of a water droplet, an insulating film, a sheet of paper with printed electrodes, a base plate, and a high-voltage control unit.

To assemble the system, a sheet of paper with freely designed electrode patterns is first placed on the base plate. Silicone oil is then applied to the paper's surface, followed by placement of the insulating film. Next, a second layer of silicone oil is applied on the insulating film to create a slippery surface, and the water droplet is placed on top. Each printed electrode channel on the paper is connected to the high-voltage control unit, which enables dynamic switching of the applied voltage across arbitrary electrodes. This setup allows the water droplet placed on the top layer to dynamically deform its shape in accordance with the electrode pattern.

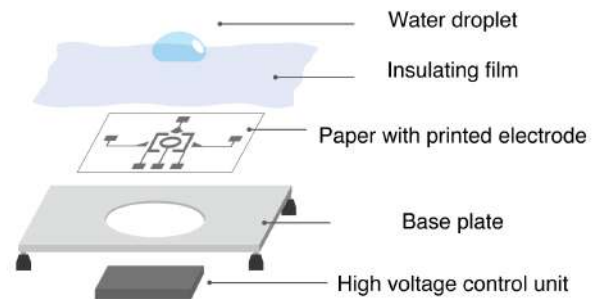


Figure 2: System design of the shape-changing water droplet setup.

2.2 Basic Principle of Shape-Changing Water Droplets

We describe the basic principle underlying the shape change of water droplets. Fig. 3 illustrates the positional relationship between a water droplet and electrodes as viewed from above. In this approach, we assume that the droplet volume is sufficiently larger than the electrode width. As shown in Fig. 3, droplet shape change can be classified into two patterns when simplified to the most basic configuration. The first pattern (Fig. 3A) corresponds to a state in which the droplet is not surrounded by electrodes to which voltage is applied. In this case, the droplet is attracted toward the voltage-applied electrode geometry and deforms by elongating along the electrode shape (linear in this example). The second pattern (Fig. 3B) corresponds to a state in which the droplet is surrounded by electrodes to which voltage is applied. In this case, the droplet is stretched along the edges of the surrounding voltage-applied electrodes, and its contour deforms accordingly.

Fig. 4 shows a side view of the droplet behavior in the state illustrated in Fig. 3B. First, in the initial state (Fig. 4A), the water droplet is positioned on the central electrode between two freely shaped electrodes. At this point, the droplet is not far from the

side electrodes, which are arranged to surround the droplet. The electrode geometry is arbitrary and can be designed according to the intended deformation of the droplet. Next, in (Fig. 4B), a high voltage (+HV) is applied simultaneously to both side electrodes. This alters the contact angle of the droplet, drawing it toward the edges of the activated electrodes. As a result, the droplet spreads in a flattened state along the shape of the electrodes, and its contour dynamically changes according to the electrode geometry. Finally, in (Fig. 4C), when the high-voltage application to both side electrodes is stopped and returned to (GND), the contact angle of the droplet reverts to its original state. The droplet then contracts toward the center and returns to its original rounded shape.

In this way, by selectively using or combining the state in which the droplet is not surrounded by voltage-applied electrodes (Fig. 3A) and the state in which it is surrounded by voltage-applied electrodes (Fig. 3B), droplet shape changes corresponding to the electrode arrangement can be achieved through control of these states.

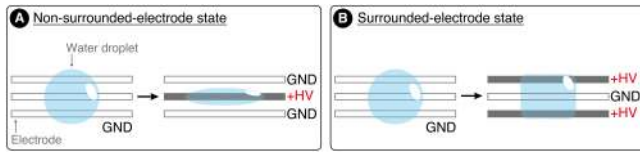


Figure 3: Basic principle of shape-changing water droplets viewed from above. (A) A non-surrounded-electrode state, where the droplet elongates along the voltage-applied electrode geometry. (B) A surrounded-electrode state, where the droplet deforms along the edges of surrounding voltage-applied electrodes.

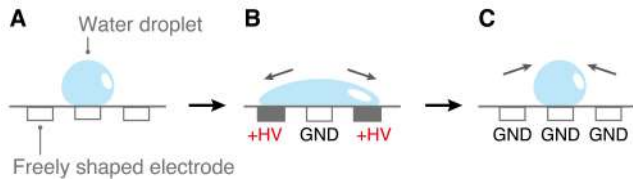


Figure 4: Side view of shape-changing water droplets in the surrounded-electrode state. (A) A water droplet is positioned between two freely shaped electrodes on the sides. (B) When high voltage is applied to both side electrodes, the droplet is pulled toward the edges and spreads along the electrode shape. (C) After turning off the high-voltage, the droplet contracts and returns to its original rounded form.

2.3 Dynamic Control of Multiple Shape-Changing Patterns

This subsection describes the mechanism for dynamically transforming a droplet into multiple shape patterns on a single sheet. Fig. 5 illustrates an example of the surrounded-electrode state, in which a droplet changes into three distinct shapes—circle, triangle, and square—along with the corresponding electrode design and the

sequence of operations. In this configuration, the electrodes printed on the paper consist of six channels.

First, as shown in Fig. 5A, when high voltage is applied only to a single circular electrode while all other electrodes are grounded, the droplet remains centered and maintains a circular shape. Next, as shown in Fig. 5B, when high voltage is simultaneously applied to three electrodes corresponding to the vertices of a triangle, and the remaining electrodes are grounded, the droplet deforms into a triangular shape. Then, in Fig. 5C, by applying high voltage to two pairs of electrodes positioned at the four corners of a square and grounding the rest, the droplet transforms into a square shape.

In this way, by selectively switching the electrodes to which voltage is applied, the droplet can dynamically transform into multiple shape patterns—such as a circle, triangle, and square—in any order on a single sheet.

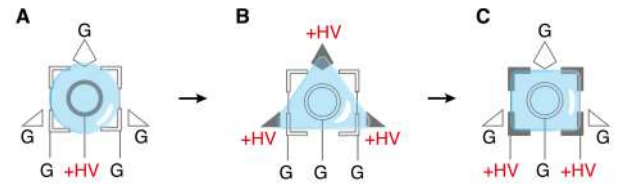


Figure 5: Mechanism for dynamically controlling multiple shape-changing patterns. (A) Applying high voltage to a circular electrode keeps the droplet centered in a circular shape. (B) Applying high voltage to three electrodes in a triangular pattern transforms the droplet into a triangle. (C) Applying high voltage to four corner electrodes transforms the droplet into a square.

3 Design Examples

This work allows electrodes to be easily printed, enabling a wide range of designs to be explored. We present several design examples that leverage the unique features of the approach by combining custom electrode patterns with multi-electrode voltage control.

3.1 Circle, Triangle, and Square Patterns

Fig. 6 shows an example incorporating circular, triangular, and square electrode patterns in the surrounded-electrode state. Six electrode channels are arranged on a single sheet, each designed to form one of the three shapes. The voltage applied to the electrodes corresponding to each shape is switched at one-second intervals, causing the droplet to dynamically transform into a circle, triangle, or square along the contours of the activated electrode patterns.



Figure 6: The behavior of a shape-changing water droplet on circular, triangular, and square electrode patterns in the surrounded-electrode state.

3.2 Radial Pattern

Fig. 7 shows an example using a radial pattern in the non-surrounded-electrode state. Twelve linear electrode channels are arranged radially at equal intervals on a single sheet, designed so that the electrodes do not make contact at the center. The voltage applied to each electrode is switched sequentially at one-second intervals, producing dynamic shape changes reminiscent of the movement of a clock's hands.

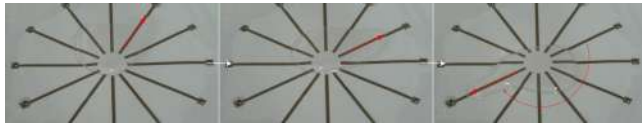


Figure 7: The behavior of a shape-changing water droplet on a radial electrode pattern in the non-surrounded-electrode state.

3.3 Sparsely Spaced Parallel Pattern

Fig. 8 illustrates an example using a sparsely spaced parallel pattern in the non-surrounded-electrode state. Although the droplet appears to be surrounded by multiple electrodes, voltage is applied to only a single electrode at a time, resulting in behavior corresponding to the non-surrounded-electrode state. Ten linear electrode channels are arranged in parallel at equal intervals with wide spacing. The voltage applied to each electrode is switched sequentially at one-second intervals, producing amoeba-like shape changes in which the large droplet is repeatedly stretched and shrunk along the electrodes.



Figure 8: The behavior of a shape-changing water droplet on a sparsely spaced parallel electrode pattern in the non-surrounded-electrode state. Voltage is applied to only a single electrode at a time, resulting in non-surrounded-electrode behavior.

3.4 Closely Spaced Parallel Pattern

Fig. 9 shows an example using a closely spaced parallel pattern in the surrounded-electrode state. Ten linear electrode channels are arranged in parallel at equal intervals with compact spacing. The voltage applied to each electrode is switched sequentially at one-second intervals, first from the center outward and then from the periphery inward. During this process, the large droplet is repeatedly stretched and shrunk along the electrodes, and because voltage is applied to multiple electrodes simultaneously, the droplet undergoes separation and merging, producing shape changes reminiscent of wave propagation.

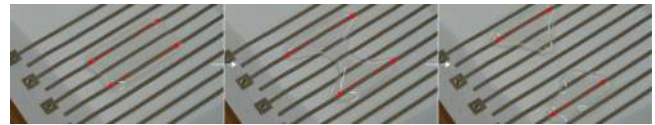


Figure 9: The behavior of a shape-changing water droplet on a closely spaced parallel pattern in the surrounded-electrode state.

4 Conclusion

In this demonstration, we present a novel approach for enabling dynamic shape-changing of water droplets on conductive patterns printed on a paper-like sheet. This work supports easy customization of electrode shapes and layouts using a household inkjet printer and commercially available silver nanoparticle ink, enabling droplet shape deformation through wettability control. By switching the activated electrodes, multiple patterns of shape-changing droplets can be dynamically realized on a single sheet. We also present several design examples that demonstrate the expressive potential of this approach. As a next step, we plan to investigate the limitations of droplet shape transformation through evaluation, develop intuitive design support tools that allow users to easily create shape-changing droplet patterns, and explore practical use cases based on the presented design examples. We believe that this approach contributes to the expressive capacity of droplet-based interfaces and suggests new directions for interaction design.

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